# Introduction to algorithm design

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*Lecture 7*

# Outline

- Algorithm design strategies
  - Brute force
  - Divide-and-conquer
  - Dynamic programming

# Algorithm design strategies

- "Templates" for creating algorithms
- Need to be adapted to specific problem
- Give you more tools for approaching difficult problems

- **Strategy 0:** brute force
  - A.k.a., exhaustive search
  - Test all possibilities for the solution
  - Report the correct/best solution

# Brute force example

- Brute force sorting

**Input:** $data$: array to be sorted
**Input:** $n$: length of $data$
**Output:** permutation of $data$ such that
$data[1] \leq \ldots \leq data[n]$

```
Algorithm: BruteSort
repeat
    Find the next permutation of data to test
    if data is sorted then
        return data
    end
until no more permutations of data are left
error data cannot be sorted
```

# Analysis: brute force

**Pros**

- Applicable to most problems
- *Always* gets the correct/optimum answer
- Easy to design and describe

**Cons**

- Almost always slowest solution
- Often infeasible
- Exponential or factorial number of tests are impractical for most realistic problems

# Divide-and-conquer

- **Goal**
  - Reduce complexity of high-complexity algorithms

- **Outline**
  - Divide large problems into one or more subproblems of roughly the same size
    - E.g., split array into 2 halves, 3 thirds, etc.
  - Solve subproblems via recursion
  - Combine solutions to subproblems into solution for full problem
  - Solve small problems directly (*base case*)

- **Intuition**
  - If combining solutions is easier than solving directly, divide-and-conquer solution may be faster

# Divide-and-conquer examples

- General strategy
  - Divide problem into equal parts
  - Solve subproblems recursively
  - Combine solutions
  - Solve small instances directly

- MergeSort
  - Split array in half
  - Recursively sort each half
  - Merge sorted arrays
  - Base case: 0 or 1 element

$2T(n/2)$

$\Theta(n)$

$T(n) = 2T(n/2) + \Theta(n)$

$T(n) = \Theta(n \lg n)$

# Searching

- **Input**
  - *data*: sorted array of length *n*
  - *t*: target value
- **Output:** index *i* such that *data[i]* = *t*, or -1 if $t \notin data$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | *t* | | | |

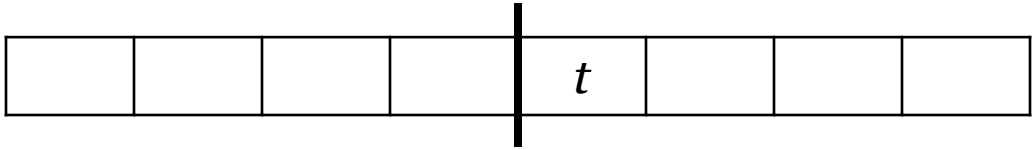
- Brute force
  - Check all values for *t*
  - Return when found
  - Worst-case: O(*n*)
- *Optimization:* stop searching if *data[i]* > *t*
  - Doesn't improve worst case

**Input:** $data$: sorted array to search
**Input:** $n$: length of $data$
**Input:** $t$: target value
**Output:** Index $i$ such that $data[i] = t$, or $-1$ if $t \notin data$

```
Algorithm: ExhaustiveSearch
for i = 1 to n do
    if data[i] = t then
        return i
    end
end
return −1
```

# Naïve binary search

- **Input**
  - *data*: sorted array of length *n*
  - *t*: target value
- **Output:** index *i* such that *data[i]* = *t*, or -1 if $t \notin data$




- Split *data* in half
- Search each half
- Return *t* if found
- Base case: array size 1

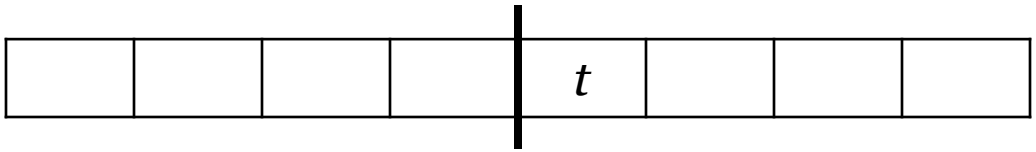
- $T(n) = 2T(n/2) + \Theta(1)$
  - $T(n) = \Theta(n)$ by Master Theorem
  - No better than linear search!

**Input:** $data$: sorted array to search
**Input:** $n$: length of $data$
**Input:** $t$: target value
**Output:** Index $i$ such that $data[i] = t$, or $-1$ if $t \notin data$

```
Algorithm: NaïveBinSearch
if n = 1 then
    if data[1] = t then
        return 1
    else
        return −1
    end
end
mid = ⌊n/2⌋
lhs = NaïveBinSearch(data[1..mid])
rhs = NaïveBinSearch(data[mid + 1..n])
if lhs ≠ −1 then
    return lhs
else if rhs ≠ −1 then
    return mid + rhs
else
    return −1
end
```

# Better binary search

- **Input**
  - *data*: sorted array of length $n$
  - $t$: target value
- **Output:** index $i$ such that $data[i] = t$, or -1 if $t \notin data$




- *Observation*
  - In sorted array, only one half has $t$

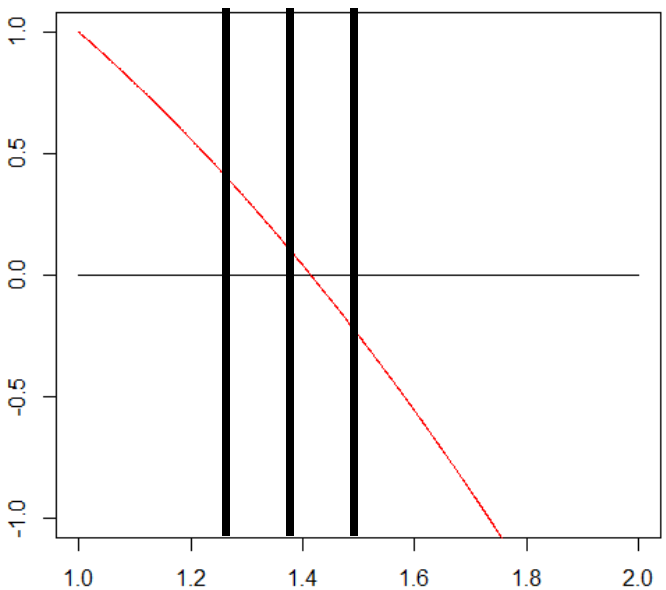
- Reduces number of recursive calls to 1
  - $T(n) = T(n/2) + \Theta(1)$
  - $T(n) = \Theta(\lg n)$ by MT

**Input:** $data$: sorted array to search
**Input:** $n$: length of $data$
**Input:** $t$: target value
**Output:** Index $i$ such that $data[i] = t$, or $-1$ if $t \notin data$

```
Algorithm: BinSearch
if n = 1 then
    if data[1] = t then
        return 1
    else
        return −1
    end
end
mid = ⌊n/2⌋
if data[mid] = t then
    return mid
else if data[mid] > t then
    return BinSearch(data[1..mid − 1])
else
    return mid + BinSearch(data[mid + 1..n])
end
```

# One-sided binary search

- Binary search can be applied to many different types of problems
  - E.g., estimating a root of $2-n^2$ between 1 and 2
    - Guess 1.5 (too high)
    - Guess 1.25 (too low)
    - Guess 1.375 (too low)
    - …
    - Repeat until sufficiently close
- Needs upper and lower bounds




- Alternative: one-sided search $f(1.25) = 0.4375$ $f(1.5) = -0.25$
  - Start with min value
  - Double until too large
  - Binary search between last success and first failure

# One-sided binary search example

- Compute $\lfloor \lg n \rfloor$ for positive $n$
  - Lower bound?
  - Upper bound?

- Exhaustive solution
  - Try every power of 2 until you exceed $n$
  - $\Theta(\lg n)$ time

- One-sided binary search
  - *Double* power of 2 until too high
  - Binary search between last success and first failure
  - $\Theta(\lg \lg n)$ time

```
Input: n: positive integer
Output: ⌊lg n⌋
Algorithm: LinearScanLog
f = 1
x = 0
while f < n do
    f = 2f
    x = x + 1
end
return x
```

```
Input: n: positive integer
Output: ⌊lg n⌋
Algorithm: OneSidedLog
lo = 0
hi = 1
while 2^hi < n do
    lo = hi
    hi = 2hi
end
while hi − lo > 1 do
    mid = ⌊(hi + lo)/2⌋
    if 2^mid < n then
        lo = mid
    else
        hi = mid
    end
end
return lo
```

# Divide-and-conquer analysis

- **Pros**
  - Reduces complexity for several problems
  - Easy to prove correctness via strong induction
  - Easy to analyze with Master Theorem
  - Works very well with parallel computing

- **Cons**
  - Problem must exhibit *optimal substructure*
    - Solution must be related to subproblem solutions
  - Sometimes has poor complexity even with optimal substructure
    - *Dynamic programming* solves this problem

# Divide-and-conquer exercise

- Develop an efficient algorithm for *natural number exponentiation*
- **Input**
  - $a$: base of exponent (real number)
  - $n$: exponent (nonnegative integer)
- **Output:** $a^n$

- Divide problem into equal parts
- Solve subproblems recursively
- Combine solutions
- Solve small instances directly

- *Hint:* $a^n = \underbrace{a \cdot a \cdot a \cdots a}_{n \text{ times}}$
  - Focus on even $n$ first
  - Think about how to handle odd $n$

# Divide-and-conquer application

- Consider matrix addition and multiplication:
  - Given $n$-by-$n$ matrices $A$ and $B$, compute the sum $C = A + B$
  - Given an $n$-by-$n$ matrices $A$ and $B$, compute the product $C = AB$
- Addition
  - Compute $c_{ij} = a_{ij} + b_{ij}$, for every $cij$
  - Complexity: $\Theta(1)\Theta(n^2) = \Theta(n^2)$
- Multiplication
  - Compute $c_{ij}$ = sum-product of row $i$ of A and column $j$ of $B$
    - $c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$
  - Complexity: $\Theta(n)\Theta(n^2) = \Theta(n^3)$
  - Fastest known algorithm until 1969

# Strassen's Algorithm

- Described by Volker Strassen (1969)
- Used divide-and-conquer to reduce complexity of matrix multiplication
- Divide matrices into 4 quadrants:

$$\begin{array}{|c|c|}\hline A_{11} & A_{12} \\ \hline A_{21} & A_{22} \\ \hline \end{array} \times \begin{array}{|c|c|}\hline B_{11} & B_{12} \\ \hline B_{21} & B_{22} \\ \hline \end{array} = \begin{array}{|c|c|}\hline C_{11} & C_{12} \\ \hline C_{21} & C_{22} \\ \hline \end{array}$$

- Compute the following:

$$M_1 = (A_{11} + A_{22})(B_{11} + B_{22})$$
$$M_2 = (A_{21} + A_{22})B_{11}$$
$$M_3 = A_{11}(B_{12} - B_{22})$$
$$M_4 = A_{22}(B_{21} - B_{11})$$
$$M_5 = (A_{11} + A_{12})B_{22}$$
$$M_6 = (A_{21} - A_{11})(B_{11} + B_{12})$$
$$M_7 = (A_{12} - A_{22})(B_{21} + B_{22})$$

- Then:

$$C_{11} = M_1 + M_4 - M_5 + M_7$$
$$C_{12} = M_3 + M_5$$
$$C_{21} = M_2 + M_4$$
$$C_{22} = M_1 - M_2 + M_3 + M_6$$

# Strassen's complexity

- Strassen's equations:

$$M_1 = (A_{11} + A_{22})(B_{11} + B_{22})$$
$$M_2 = (A_{21} + A_{22})B_{11}$$
$$M_3 = A_{11}(B_{12} - B_{22})$$
$$M_4 = A_{22}(B_{21} - B_{11})$$
$$M_5 = (A_{11} + A_{12})B_{22}$$
$$M_6 = (A_{21} - A_{11})(B_{11} + B_{12})$$
$$M_7 = (A_{12} - A_{22})(B_{21} + B_{22})$$

$$C_{11} = M_1 + M_4 - M_5 + M_7$$
$$C_{12} = M_3 + M_5$$
$$C_{21} = M_2 + M_4$$
$$C_{22} = M_1 - M_2 + M_3 + M_6$$

  - Total complexity: $T(n) = 7T(n/2) + \Theta(n^2)$
    - 18 additions/subtractions $\Theta(n^2)$
    - 7 multiplications $7T(n/2)$
- $c = \log_2(7) \approx 2.81$
- $f(n)$ vs. $n^{2.81} \Rightarrow f(n) = O(n^{\lg 7 - 0.8})$
- $T(n) = \Theta(n^c) \approx \Theta(n^{2.81})$

# Dynamic programming

- Algorithm design strategy
- Similar to divide-and-conquer
- Useful when divide-and-conquer would be inefficient

# Dynamic programming motivation

$$F(n) = \begin{cases} 1, & \text{if } n = 1, 2 \\ F(n-1) + F(n-2), & \text{if } n > 2 \end{cases}$$

```
Input: n: index of Fibonacci number to compute
Output: F_n
Algorithm: Fibonacci
if n = 1 or 2 then
    return 1
else
    return Fibonacci(n − 1) + Fibonacci(n − 2)
end
```

- You've probably been told that this function is inefficient
- **Review:** recursive complexity
- *$T(n) = T(n-1) + T(n-2) + \Theta(1)$*
- Complexity: $\Theta(\tau^n)$, where $\tau = \frac{1+\sqrt{5}}{2} \approx 1.61$
  - Calculating $F_{1000}$ takes forever!

# A better solution

| 1 | 1 | 2 | 3 | 5 | 8 | 13 |
|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 |

**Input:** $n$: index of Fibonacci number to compute
**Output:** $F_n$

```
Algorithm: FastFib
fib = Array(n)
Initialize fib to −1
return DynamicFib(n)
```

```
Algorithm: DynamicFib(n)
if fib[n] = −1 then
    if n = 1 or 2 then
        fib[n] = 1
    else
        fib[n] = DynamicFib(n − 1) +
          DynamicFib(n − 2)
    end
end
return fib[n]
```

- Each Fibonacci number computed *once*
- Total complexity: $\Theta(n)$

# Dynamic programming

- General algorithm design strategy
- Recursion augmented with data structure to save sub-results
  - Not always a 1D array…
    - Depends on number of parameters
- **Useful when:**
  - Problem has recursive structure (*like divide-and-conquer*)
  - More than one recursive call
  - Problems overlap
    - Typically, 1-2 levels deep in recursion tree
- **Related concept:** memoization
  - Function that stores sub-results
  - Returns immediately when calling with previously-seen value
  - *Note:* possible to memoize non-recursive functions
    - Dynamic programming can be implemented iteratively

# A more complex example

- Binomial coefficients ("$n$ choose $k$")
  - **Formula:** $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
  - **Example:** $\binom{20}{3} = \frac{20!}{3!(17!)}$

$$= \frac{2432902008176640000}{6(355687428096000)}$$

$$= 1140$$

- Formula might cause overflow
- Alternative: Pascal's triangle

$k \longrightarrow$

$n \downarrow$

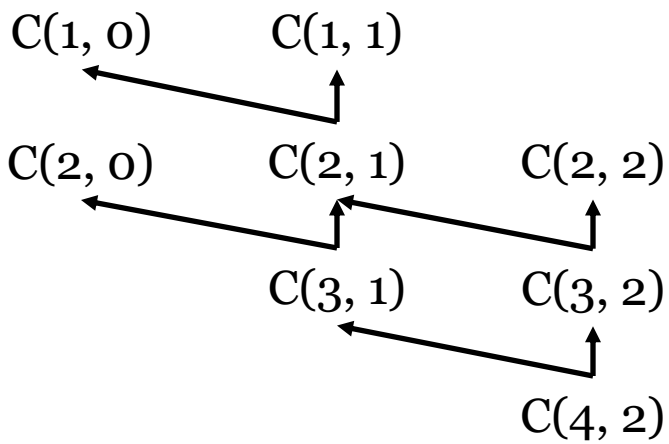
| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | 1 | | | | |
| | | | | 1 | | 1 | | | |
| | | | 1 | | 2 | | 1 | | |
| | | 1 | | 3 | | 3 | | 1 | |
| | 1 | | 4 | | 6 | | 4 | | 1 |

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

# Binomial coefficients example

- Recursive function
- **Example:** C(4, 2)

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$




- Overlapping subproblems
  - Dynamic programming!

# Memoization summary

- Five step procedure

1. **Start with naïve recursive algorithm**
   - Based on recurrence
2. **Decide data structure and sentinel value**
3. **Add "memoization check" to the beginning of the algorithm**
   - If solution is in data structure, return it
4. **Store solution before returning**
5. **Write wrapper function to initialize data structure**

$$\binom{n}{k} = \begin{cases} 1, \text{ if } k = 0 \text{ or } k = n \\ \binom{n-1}{k-1} + \binom{n-1}{k} \end{cases}$$

**Input:** $n$, $k$: binomial coefficient to compute
**Output:** $\binom{n}{k}$

```
Algorithm: DPBinom
binom = Array(n, k)
Initialize binom to 0
return MemoBinom(n, k)
```

```
Algorithm: MemoBinom
if binom[n, k] > 0 then
    return binom[n, k]
else if k = 0 or k = n then
    binom[n, k] = 1
else
    binom[n, k] = MemoBinom(n−1, k−1)
      + MemoBinom(n − 1, k)
end
return binom[n, k]
```

# Memoization exercise

- Stirling Numbers of the Second Kind, *S(n, k),* count the number of ways to divide *n* objects into *k* groups.
- They satisfy the following recurrence:

$$S(n,k) = \begin{cases} 1, & \text{if } k = 1 \text{ or } k = n \\ kS(n-1,k) + S(n-1,k-1), & \text{otherwise} \end{cases}$$

1. Give pseudocode for a naïve recursive algorithm that computes *S(n, k)* using this recurrence.
2. What map implementation would you use for a memoized dynamic programming solution to this problem and why?
3. What would be a reasonable sentinel value to represent an unsolved problem in this map?
4. Give pseudocode for a memoized dynamic programming algorithm for this problem.

# Coming up

- **Midterm Exam** next week (10/22)
  - Practice Exam and sample solution posted
  - Prepare sheet of notes for exam
  - *Optional review session:* Monday at 6pm via Teams (link)
  - **Feedback Form** due on day of exam
- **Recommended readings:** review!!!

- **<u>Post-exam</u>**
- Dynamic programming
  - Longest common subsequence
- Greedy algorithms
- *Recommended readings (post-exam):* Sections 11.3, 11.2, 12.1, 12.2, 12.5
  - *Practice problems:* R-11.3, C-11.3 (complexity only), C-11.4, A-11.2, R-12.9, C-12.6, C-12.7, C-12.9, C-12.10, A-12.1, A-12.2, A-12.5